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Siridhanya Santhe – Digital Millet Marketplace

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Outlines

1. Problem definition
2. Introduction
3. Objectives
4. Workflow
5. Technologies
6. Conclusion

Problem Statement

The Problem

Millet farmers and SHGs lack direct access to buyers

No transparent pricing or product traceability

Rural users need simple, mobile-friendly tools

Our Vision

A digital marketplace that removes middlemen

Enables direct farm-to-consumer transactions

Supports 4 user roles: Farmer, Buyer, Consumer, Admin

Introduction

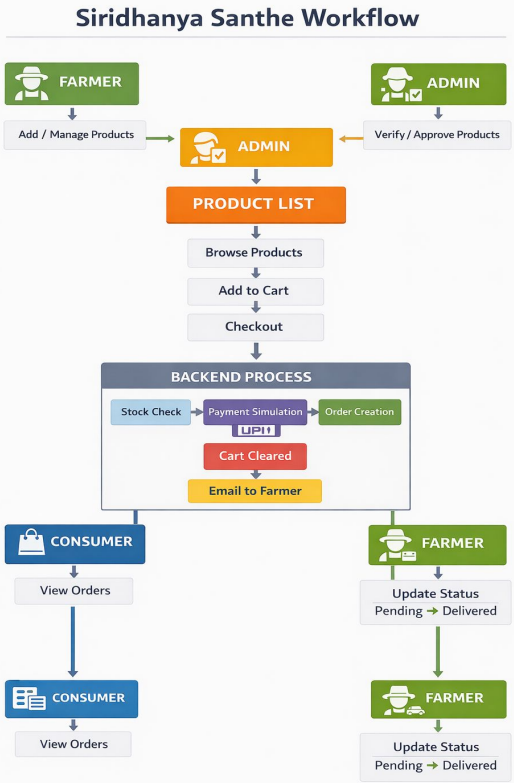
Siridhanya Santhe is a full-stack digital marketplace built specifically for millet products.

- "Siridhanya" = Millets (Kannada) | "Santhe" = Marketplace
- Connects Farmers / SHGs → Buyers → Consumers directly
- Eliminates middlemen with transparent pricing
- Supports 4 user roles: Farmer, Buyer, Consumer, Admin
- Built with the MERN Stack (MongoDB, Express, React, Node.js)
- Designed for rural accessibility — mobile-first, simple UI

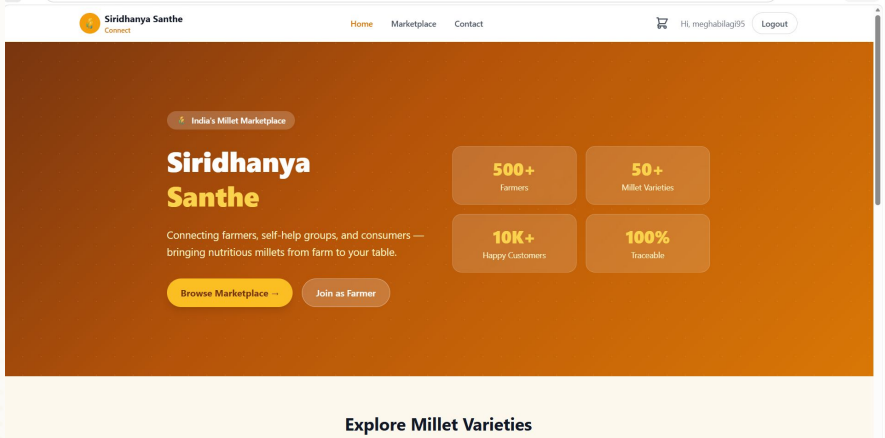
Objectives

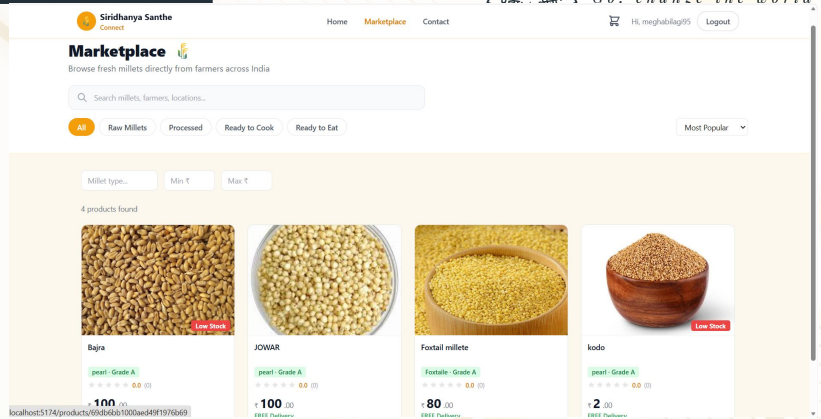
1. Let farmers list & manage products
2. Enable search, filter & purchase for buyers
3. Track orders from placement to delivery
4. Admin verifies products before going live
5. Ensure product origin traceability
6. Allow reviews from verified buy

Workflow

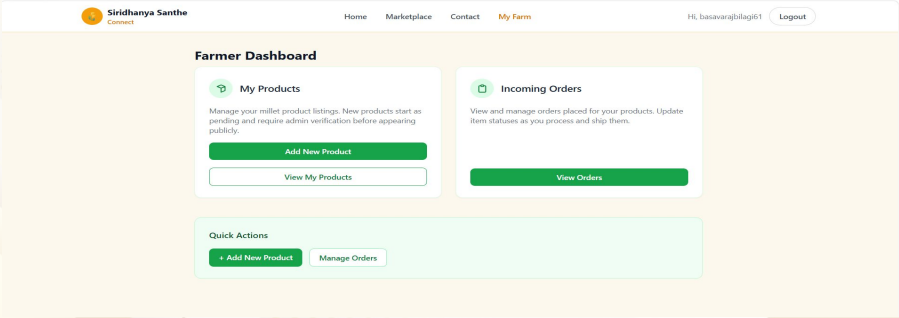

Siridhanya Santhe Workflow
FARMER
Add / Manage Products
ADMIN
Verify / Approve Products
ADMIN
PRODUCT LIST
Browse Products
Add to Cart
Checkout
BACKEND PROCESS
Stock Check
Payment Simulation
UPI
Order Creation
Cart Cleared
Email to Farmer
CONSUMER
View Orders
CONSUMER
View Orders
FARMER
Update Status
Pending → Delivered
FARMER
Update Status
Pending → Delivered

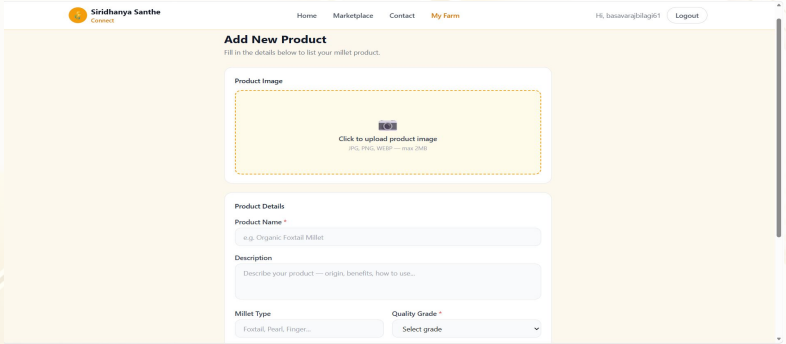
Output





Explore Millet Varieties





Output

Collection name	Properties	Storage size	Data size	Documents	Avg. document size	Indexes	Total index size
carts	-	36.86 kB	139.00 B	1	139.00 B	2	40.96 kB
orders	-	36.86 kB	9.01 kB	16	562.00 B	2	73.73 kB
products	-	708.61 kB	370.49 kB	6	61.75 kB	1	36.86 kB
reviews	-	4.10 kB	0 B	0	0 B	2	8.19 kB
users	-	36.86 kB	727.00 B	3	242.00 B	2	73.73 kB

localhost:27017 > siridhanya-santhe > products

Open MongoDB shell

Documents 6 | Aggregations | Schema | Indexes 1 | Validation

Type a query: { field: 'value' } or Generate query

Explain | Reset | Find | Options

ADD DATA | UPDATE | DELETE | EXPORT DATA | EXPORT CODE

50 | 1-6 of 6


```
_id: ObjectId('69db631099a885114455797c')
name : "Bajra"
description : "tyuiop;"
price : 100
quantity : 0
milletType : "pearl"
qualityGrade : "A"
farmer : ObjectId('69db62db99a885114455797b')
farmerName : "Basavaraj Bilagi"
verificationStatus : "verified"
isDeleted : false
averageRating : 0
reviewCount : 0
createdAt : 2026-04-12T09:17:04.937+00:00
updatedAt : 2026-04-12T10:27:43.225+00:00
__v : 0

_id: ObjectId('69db6bb1000aed49f1976b69')
name : "JOWAR"
description : "Jowar, also known as sorghum, is a nutrient-rich, gluten-free grain th…"
price : 100
quantity : 82
milletType : "pearl"
qualityGrade : "A"
farmer : ObjectId('69db62db99a885114455797b')
farmerName : "Basavaraj Bilagi"
verificationStatus : "verified"
isDeleted : false
```

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Technologies Used

- **Frontend** **React, Tailwind CSS**
- **Backend** **Node.js, Express**
- **Database** **MongoDB**
- **Auth** **JWT, bcrypt**

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Conclusion

- Empowers rural farmers with direct market access
- Transparent pricing, no middlemen
- Secure, mobile-friendly, scalable platform
- Future: Live payments, SMS alerts, Kannada language support

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Thank You